

You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

1. [2pts each] Use the graph of the quadratic function, to answer the following:

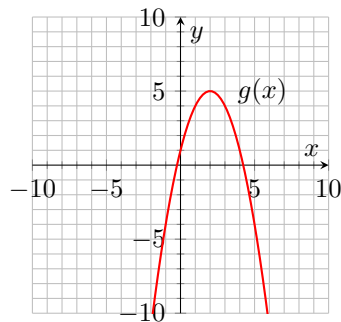
a. What is the vertex?

b. What is the axis of symmetry?

c. What is the range?

d. What is the domain?

e. Using the graph, what interval(s) is the function positive? (approximate)



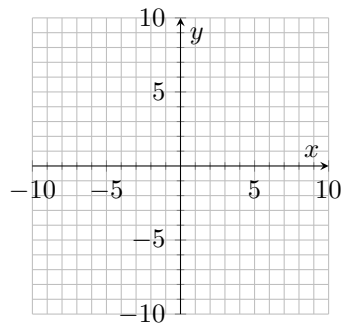
2. Using the function  $f(x) = x^2 - 4x + 4$ , complete the following:

a. [2pts] Determine  $f$ 's axis of symmetry.

b. [2pts] If any exist, find the  $x$ -int(s).

c. [2pts] Find the  $y$ -int.

d. [4pts] Sketch the graph



3. Consider  $P(x) = x^2 - 5x + 2$  and  $(x - 2)$ .

a. [4pts] Use division (synthetic or traditional) to find the quotient and remainder when  $P(x)$  is divided by  $x - 2$ .

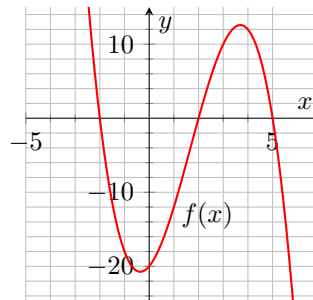
b. [4pts] Find  $\frac{P(x)}{x - 2}$

c. [2pts] Is  $x = 2$  a root/zero of  $P(x)$ ?

4. Consider the polynomial  $f(x)$  shown in the graph.

a. What are the zeros of  $f(x)$ ?

b. Find the simplest polynomial with real coefficients that has the roots from above (you do not have to expand).

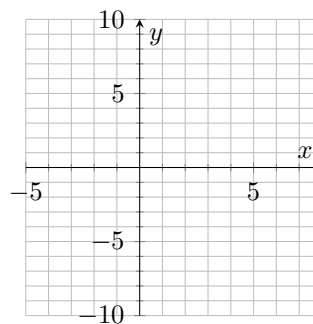


5. Use  $f(x) = x^3 - 4x$  to complete the following.

a. [3pts] Find the  $x$ -int(s).

b. [3pts] Determine if the graph crosses or touches without crossing the  $x$ -axis at each  $x$ -int.

c. [4pts] Sketch a graph of the function.



6. [5pts each] Find all real solutions.

a.  $\sqrt{x+3} = x+1$

b.  $x^{-2/5} = 4$

7. [5pts each] Find the vertex of the graph of the following functions:

a.  $f(x) = x^2 - 4x + 10$ .

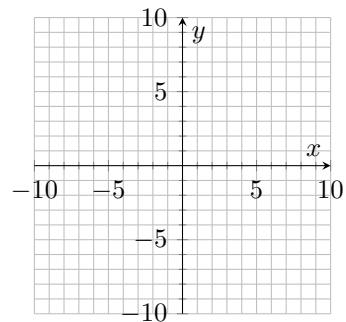
b.  $f(x) = -3(x+2)^2 - 5$ .

8. Use  $f(x) = \frac{2x+6}{x-2}$

a. [4pts] Find the domain of  $f(x)$ .

b. [4pts] Find all vertical asymptotes.

c. [6pts] Find the horizontal asymptote.



d. [6pts] Sketch a graph of the function

9. [2pts each] Find the following:

a.  $x^2 - 20$  as  $x \rightarrow -\infty$

b.  $-2x^{13} + x^4 - 50$  as  $x \rightarrow \infty$

c.  $\lim_{x \rightarrow \infty} \frac{1}{x^3}$

d.  $\lim_{x \rightarrow 0^+} \frac{3}{x}$

e.  $\lim_{x \rightarrow 2^-} \frac{3}{x-2}$

f. The domain of  $\frac{x^2 - 3x - 7}{x(x-2)(x+1)}$